

MUHAMMED RISWAN M. P.

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OBJECTIVE

Results-oriented Network Engineer with hands-on experience in SDN, 5G/O-RAN, and network automation. Proven track record of managing complex topologies across OpenStack and AWS, automating infrastructure with Python and Terraform, and optimizing routing to reduce latency and ensure 99.9% availability.

EDUCATION

MTech in Robotics and Automation, College of Engineering Trivandrum 2023 - 2025

Focus: Autonomous Systems, Mesh Communication Frameworks

BTech in Electronics and Communication, Government College of Engineering Kannur 2018 - 2022

Inter College Zonal Chess Champions (2018-2020)

SKILLS

Networking & Telecom	TCP/IP, SDN, VPC, DNS, O-RAN (CU/DU), Mesh Networks, NMS
Network Automation	Python, Bash Scripting, Ansible, Terraform, IBM Concert
Cloud & Virtualization	OpenStack (Neutron), AWS, KVM, VMware, Kubernetes (CNI)
Tools & Platforms	OpenAirInterface (OAI), Keysight Core, Git, Jenkins, Prometheus
Operating Systems	Red Hat Enterprise Linux (RHEL), Ubuntu/Debian

WORK EXPERIENCE

Tata Elxsi Bengaluru, India

Senior Engineer (Cloud & DevOps)

June 2025 - Present

Student Intern

Jan 2025 - June 2025

- **5G/O-RAN Integration:** Integrated OpenAirInterface (OAI) RAN with Keysight Core for O-RAN CU/DU architectures, achieving 99.9% protocol adherence and reducing packet routing latency by 15ms.
- **Network Management System:** Architected and deployed a custom NMS across a 20+ node KVM lab environment, improving infrastructure health monitoring and reducing network downtime by 25%.
- **Virtual Networking & SDN:** Executed live migration of 50+ virtual machines from VMware to OpenStack SDN/Neutron, resolving IP conflicts and optimizing routing to achieve zero-downtime cutovers.
- **Network Automation:** Automated network provisioning using IBM Concert, Terraform, and Ansible, reducing manual configuration errors by 40% and cutting deployment time from hours to under 15 minutes.
- **Linux Network Administration:** Resolved 100+ L3 network tickets in RHEL environments by troubleshooting TCP/IP stack issues and firewall rules, maintaining a 95% SLA compliance rate.

International Centre for Free and Open Source Solutions (ICFOSS)

Trivandrum, India

Research Intern

June 2024 - Jan 2025

- **Mesh Communication:** Developed decentralized Python mesh networking algorithms to coordinate a swarm of 10+ autonomous UAVs, increasing data transmission reliability by 30% in disaster scenarios.
- **Data Transmission:** Optimized payload delivery and telemetry streaming protocols, reducing bandwidth consumption by 20% in high-latency environments during field testing.

PROJECTS & PUBLICATIONS

Paper Publication (ETET 2025): "Scalable and Secure Mesh Communication Framework for UAV Swarm Coordination in Disaster Response". Converted from IEEE to Grenzee Journal format, highlighting robust network topology design for autonomous systems.